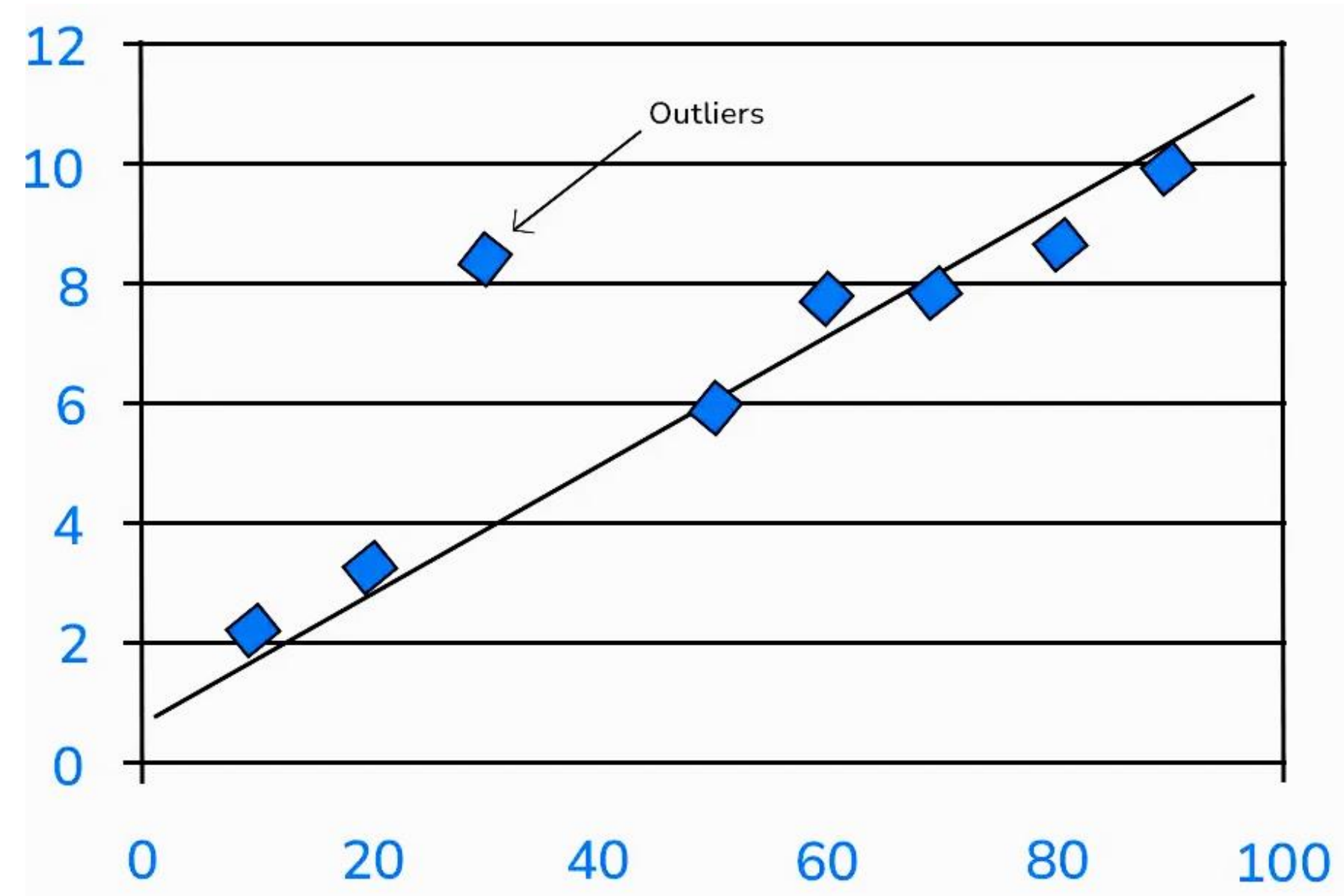


Bias-Variance Trade Off

Machine Learning Basics

Outliers

- Outliers are values thought to be out of range.
 - “An outlier is an observation that deviates so much from other observations as to arouse suspicion that it was generated by a different mechanism”
 - Can be detected by standardizing observations and label the standardized values outside a predetermined bound as outliers
 - Outlier detection can be used for fraud detection or data cleaning



Underfitting Model

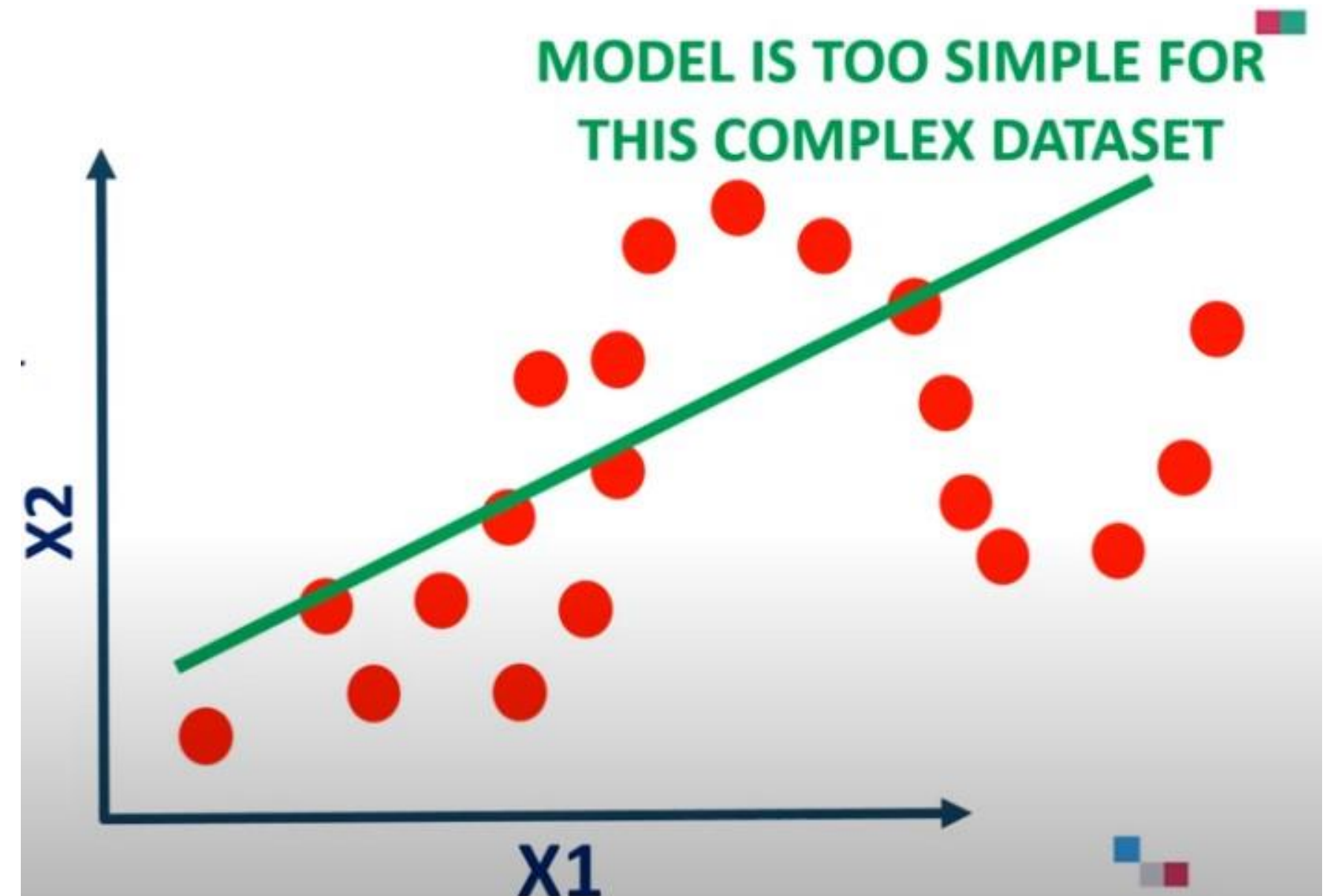
The model is under fitting if it's too simple that it cannot reflect the complexity of the training dataset.

We can overcome under fitting by:

- Increasing the complexity of the model
- Training the model for a longer period of time (more epochs) to reduce error

Here we are trying to fit the **non-linear data with Linear model**, which shows **underfitting**.

Ex: asking a car driver to drive a truck

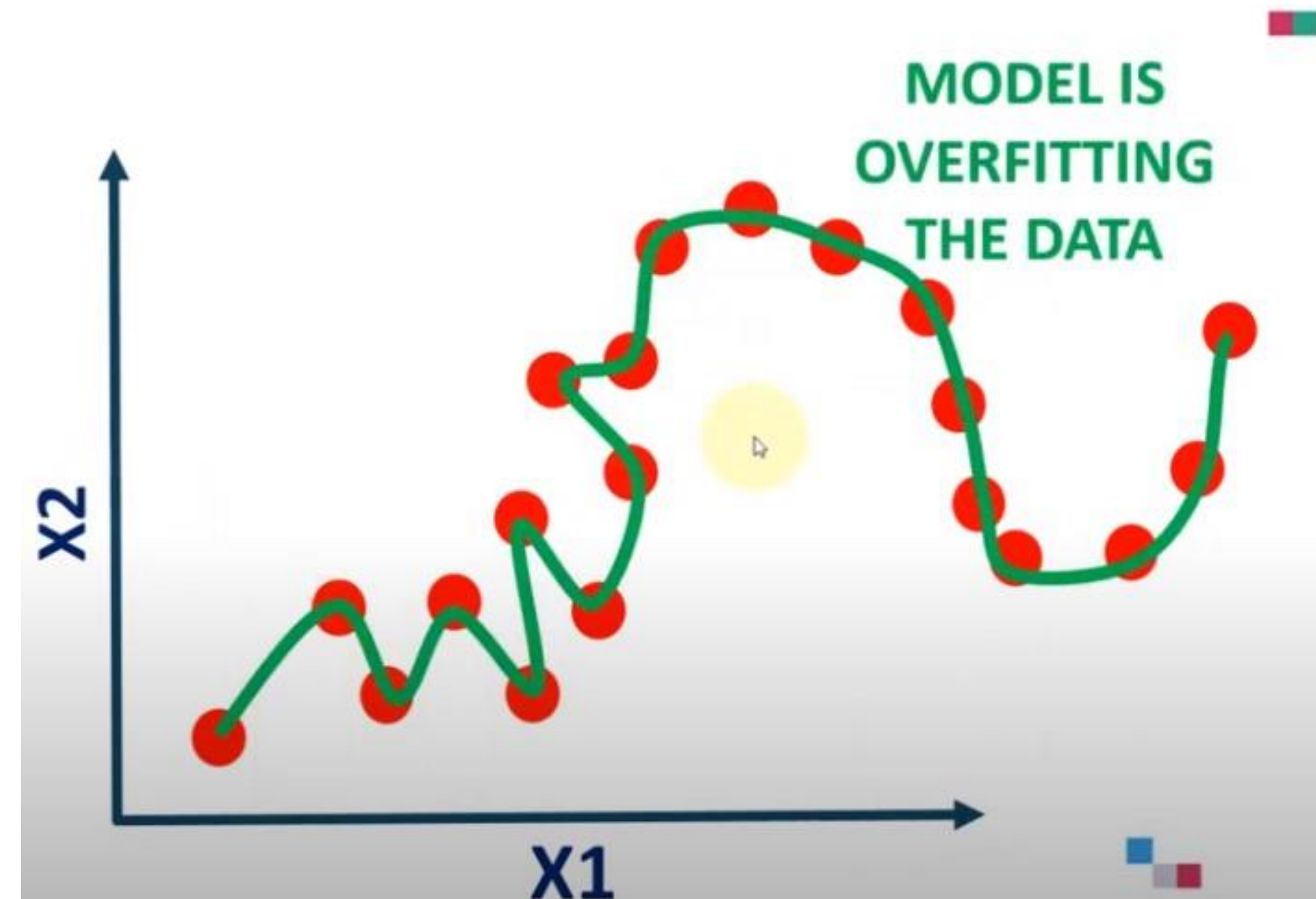


Overfitting Model

Model is overfitting data when it memorizes all the specific details of the training data and fails to generalize

- Overfitting models tends to perform very well on the training dataset but poorly on any new datasets (testing data).
- Machine learning is the art of creating models that are able to generalize and avoid memorization.

Overfitting → Poor Generalization



Overfitting:

- Overfitting occurs when a machine learning model performs very well on the training data but fails to generalize to unseen data.
- It happens when a model becomes too complex and starts to memorize the noise and outliers present in the training data, rather than learning the underlying patterns.
- As a result, the model becomes overly sensitive to the training data and fails to make accurate predictions on new instances.

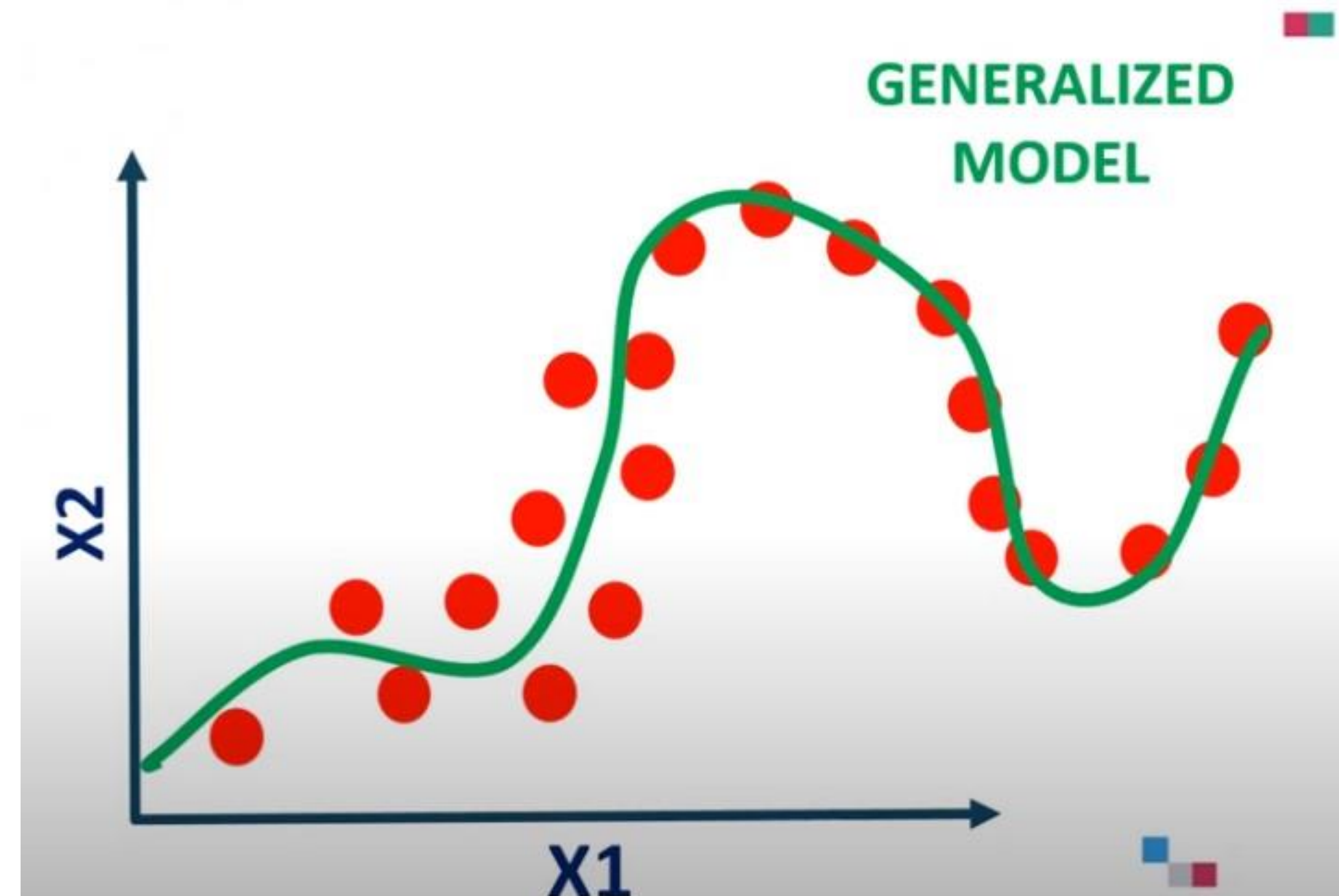
Signs of overfitting include:

- Very low or zero training error but high test error.
- Model capturing noise or outliers in the training data.
- Overfitting can be caused by various factors, such as using a complex model with too many parameters, having insufficient or noisy training data, or not using proper regularization techniques.

Generalized Model

Model that performs well during training (on new dataset that have never seen before) is considered the best model (goal)

- Generalization refers to the ability of a machine learning model to perform well on unseen or new data that it hasn't been trained on. In other words, a model with good generalization is able to learn and capture underlying patterns, trends, and relationships from the training data and apply them to make accurate predictions on new, unseen instances.
- The goal of machine learning is to build models that generalize well.



Bias-Variance Trade Off – Machine Learning

- Bias refers to error due to overly simplistic assumption in the learning algorithm.
- Bias always deals with Training Data.
- **High Bias:** When the model is too simple, it underfit the data. This means the model does not capture important patterns in the data, leading to poor performance on both the training and testing sets.
- **Low Bias:** When the model is sufficiently complex and can capture the true relationship between input and output.

What is Bias?

- The bias is known as the difference between the prediction of the values by the Machine Learning model and the correct value.
- Being high in biasing gives a large error in training as well as testing data. It is recommended that an algorithm should always be low-biased to avoid the problem of **underfitting**.
- By high bias, the data predicted is in a straight line format, thus not fitting accurately in the data in the data set. Such fitting is known as the **Underfitting of Data**.
- This happens when the hypothesis is too simple or linear in nature.

- Variance refers to **error** due to the model's sensitivity to small fluctuations in the training data.
- **High Variance:** When the model is too complex, it overfits the data. This means the model performs well on training set but poorly on new, unseen training data because it has memorized the training data rather than learning general pattern.
- **Low Variance:** When the model is general enough to perform consistently across both training and testing data.

Bias-Variance Trade Off – Machine Learning

- It is important to understand prediction errors (bias and variance) when it comes to accuracy in any machine-learning algorithm.
- There is a tradeoff between a model's ability to minimize bias and variance which is referred to as the best solution for selecting a value of **Regularization** constant.
- A proper understanding of these errors would help to avoid the overfitting and underfitting of a data set while training the algorithm.

A model has either of the two situations:

- **Low bias** – Low bias value implies fewer assumptions have been made to build the target function. In this scenario, the model will closely match the training dataset.
- **High bias** – High bias value implies more assumptions have been made to build the target function. In this scenario, the model will not match the dataset closely.

A high-bias model will be unable to capture the dataset trend. It has a high error rate and is considered an underfitting model. This happens because of a very simplified algorithm. For instance, a linear regression model might be biased if the data has a [non-linear relationship](#).

Ways To Reduce High Bias

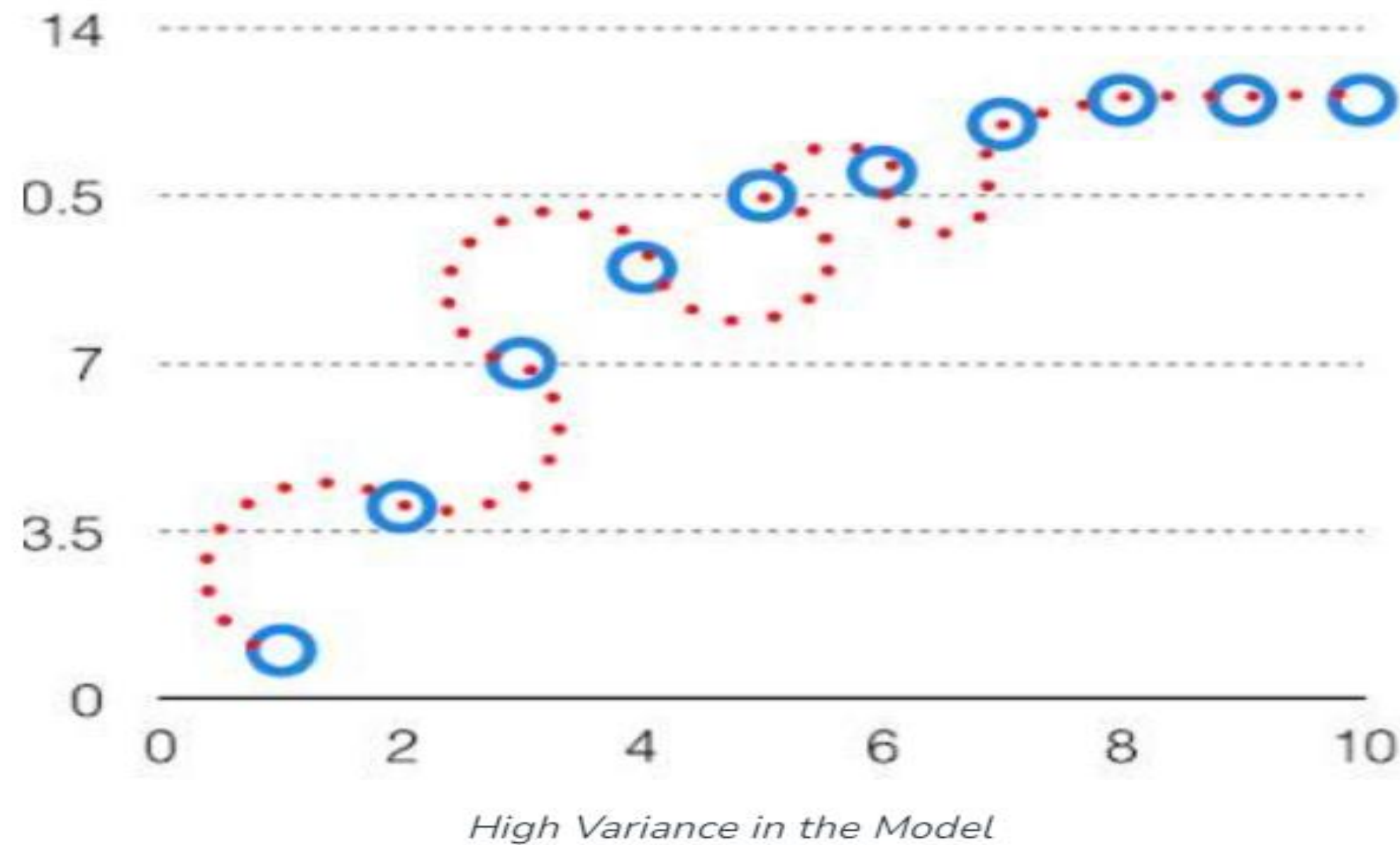
Since we have discussed some disadvantages of having high bias, here are some ways to reduce high bias in machine learning.

- **Use a complex model:** The extremely simplified model is the main cause of high bias. It is incapable of capturing the data complexity. In such scenarios, the model can be made more complex.
- **Increase the training data size:** Increasing the training data size can help reduce bias. This is because the model is being provided with more examples to learn from the dataset.
- **Increase the features:** Increasing the number of features will increase the complexity of the model. This improves the ability of the model to capture the underlying data patterns.
- **Reduce regularization of the model:** L1 and L2 regularization can help prevent overfitting and improve the model's generalization ability. Reducing the regularization or removing it completely can help improve the performance.

What is Variance?

- The variability of model prediction for a given data point which tells us the spread of our data is called the variance of the model.
- The model with high variance has a very complex fit to the training data and thus is not able to fit accurately on the data which it hasn't seen before.
- As a result, such models perform very well on training data but have high error rates on test data.
- When a model is high on variance, it is then said to as **Overfitting of Data**.
- Overfitting is fitting the training set accurately via complex curve and high order hypothesis but is not the solution as the error with unseen data is high.
- While training a data model variance should be kept low.

The high variance data looks as follows.



In such a problem, a hypothesis looks like follows.

$$h_{\theta}(x) = g(\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4)$$

Variance error is either low or high:

- **Low variance:** Low variance implies that the ML model is less sensitive to changes in the training data. The model will be able to produce consistent estimates for the target function using different data subsets of the same distribution. This is underfitting, where the model can't generalise on test and training data.
- **High variance:** High variance implies that the ML model is susceptible to changes in the training data. When trained on various subsets of data from the same distribution, the ML model can significantly change the target function estimation. This scenario is known as overfitting when the ML model does well on the training data but not on any new data.

Ways To Reduce High Variance

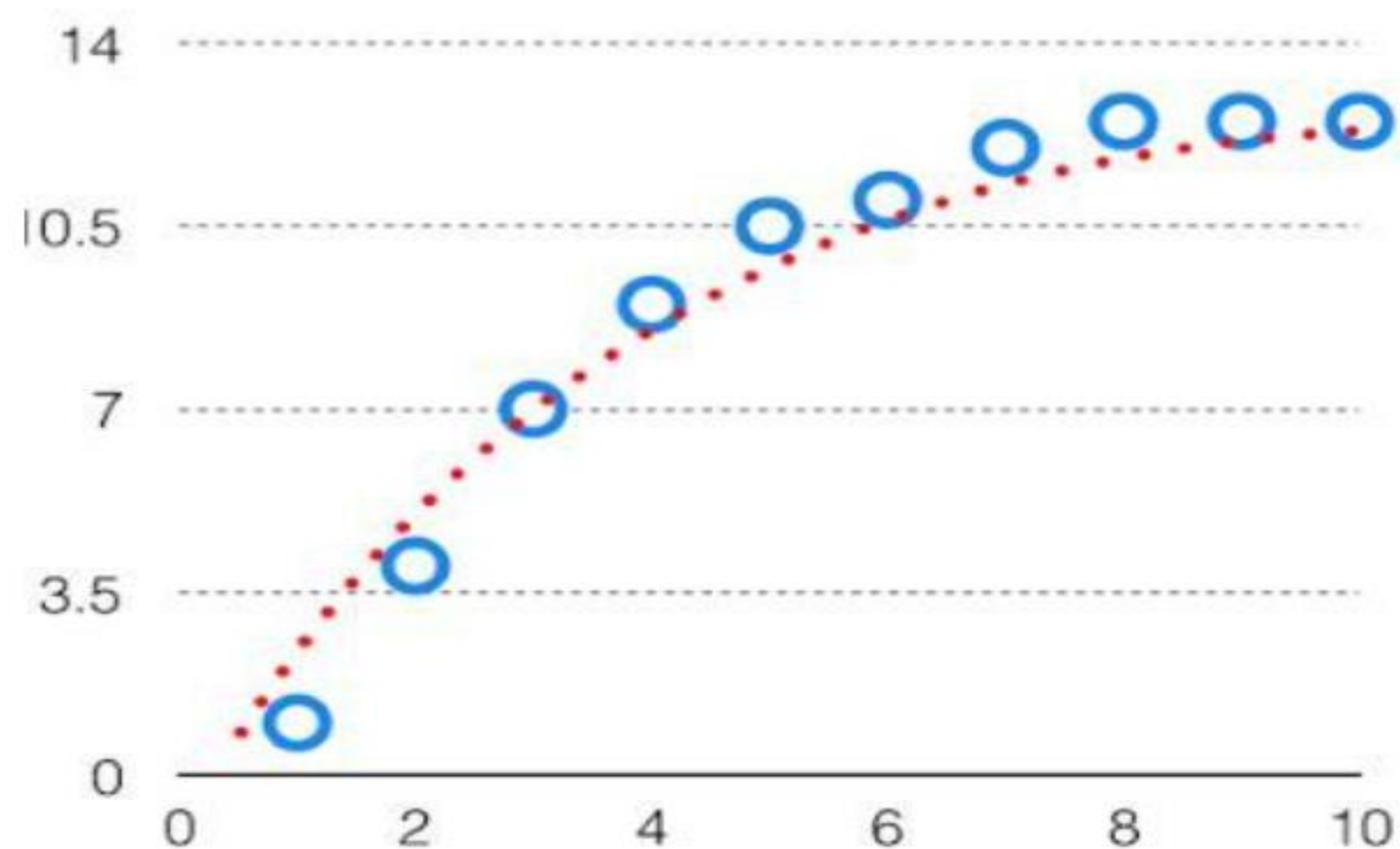
Here are some ways high variance can be reduced:

- **Feature selection:** The variance error of a model can be reduced by selecting the only relevant feature. This will decrease the complexity of the model.
- **Cross-validation:** By dividing the dataset into testing and training sets several times, cross-validation can identify if a model is underfitting or overfitting. This can be used for reducing variance by tuning the hyperparameters.
- **Simplifying the model:** Decreasing the number of parameters of neural network layers can help reduce the complexity of the model. This, in turn, helps in reducing the variance of the model.
- **Ensemble methods:** Boosting, stacking and bagging are common ensemble techniques that can help reduce the variance of an ML model and improve the generalization performance.
- **Early stopping:** This is a technique used for preventing overfitting by putting a stop to the deep learning model training when the validation set performance stops improving.

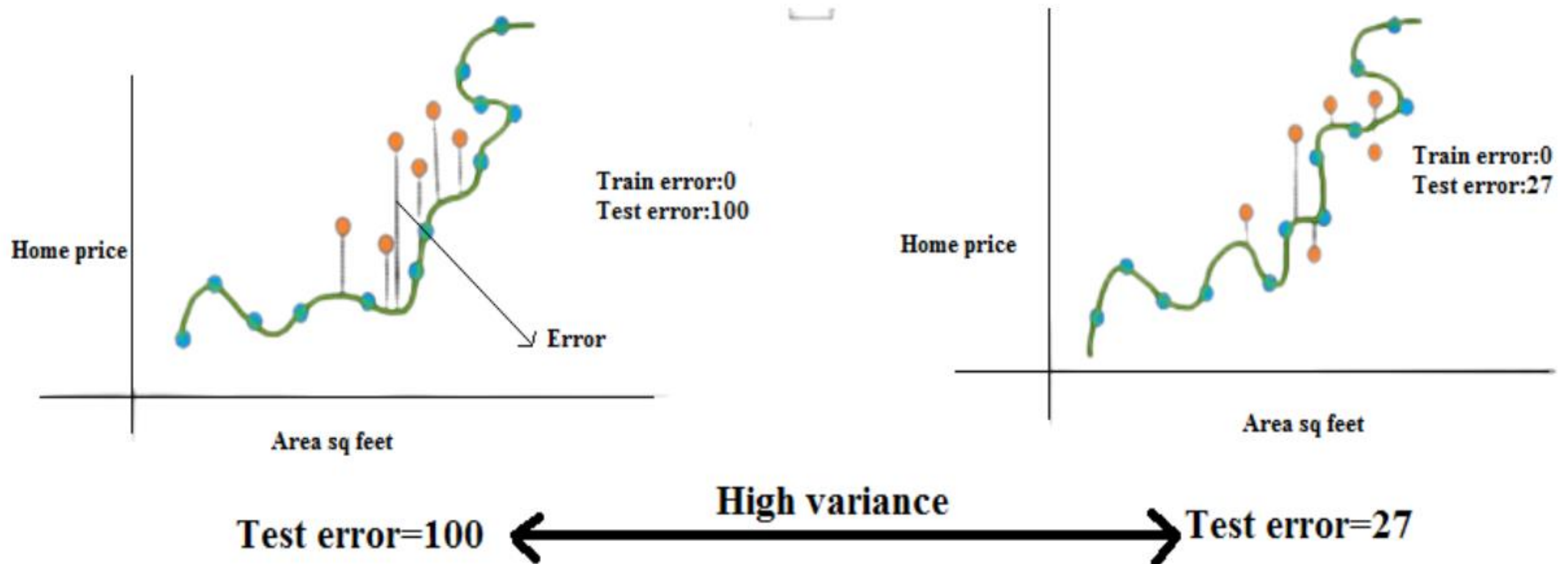
Bias Variance Tradeoff

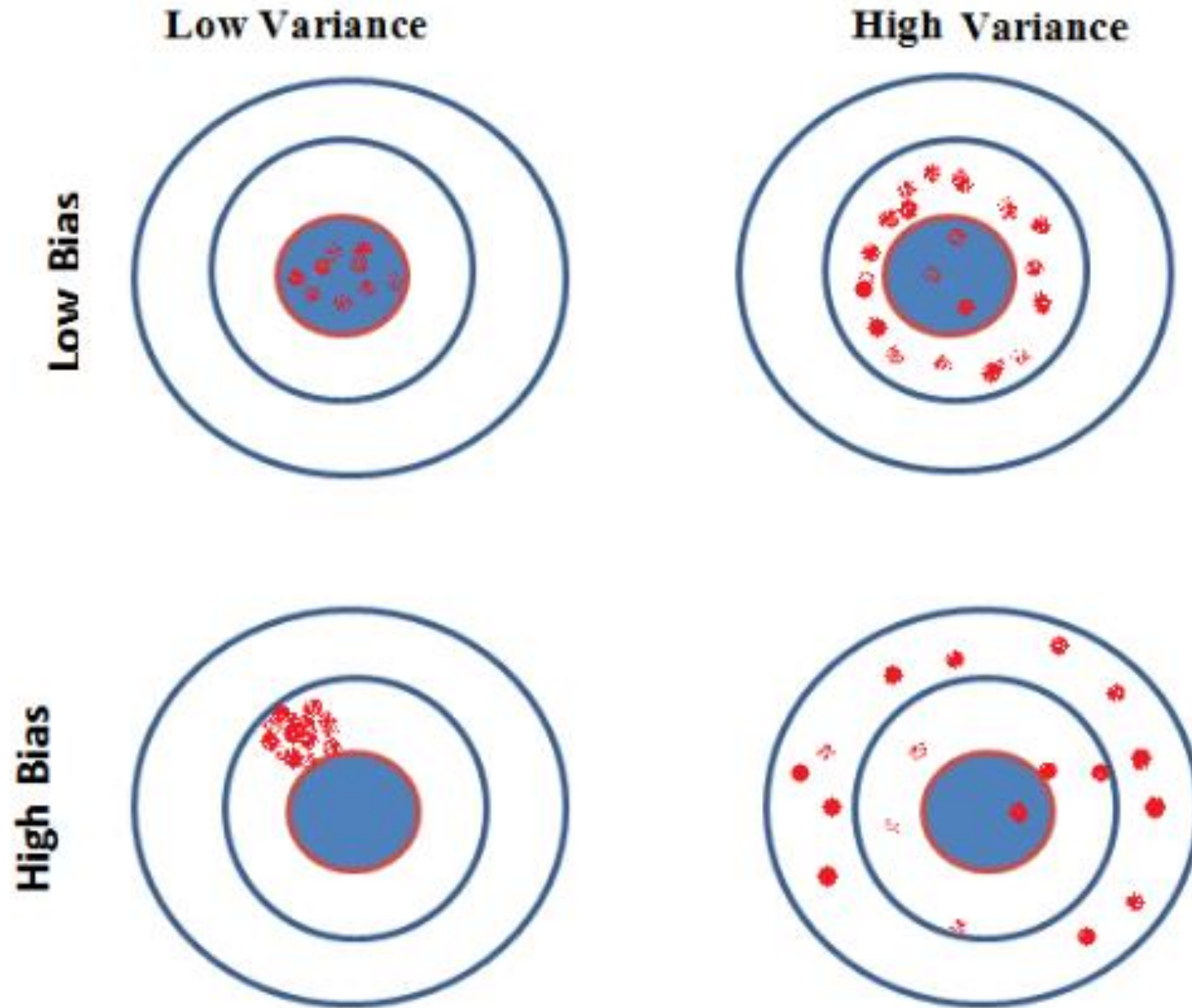
- If the algorithm is too simple (hypothesis with linear equation) then it may be on high bias and low variance condition and thus is **error-prone**.
- If algorithms fit too complex (hypothesis with high degree equation) then it may be on high variance and low bias.
- In the latter condition, the new entries will not perform well. Well, there is something between both of these conditions, known as a Trade-off or Bias Variance Trade-off.
- This tradeoff in complexity is why there is a tradeoff between bias and variance. An algorithm can't be more complex and less complex at the same time.

For the graph, the perfect tradeoff will be like this.



Bias—→Underfitting—→High train and test error
Variance—→Overfitting—→High test error





- 1.Low-Bias, Low-Variance:** The combination is an ideal machine learning model. However, it is not possible practically.
- 2.Low-Bias, High-Variance:** This is a case of **overfitting** where model predictions are inconsistent and accurate on average. The predicted values will be accurate(average) but will be scattered.
- 3.High-Bias, Low-Variance:** This is a case of **underfitting** where predictions are consistent but inaccurate on average. The predicted values will be inaccurate but will be not scattered.
- 4.High-Bias, High-Variance:** With high bias and high variance, predictions are inconsistent and also inaccurate on average.

Various Combinations of Bias-Variance

Combination	Characteristic
High bias, low variance	This type of model is said to be underfitting.
High variance, low bias	This type of model is said to be overfitting.
High bias, high variance	This model cannot capture underlying patterns in the dataset (high bias) and is too sensitive to the training data changes (high variance). Due to this, the model will mostly give inaccurate and inconsistent predictions.
Low bias, low variance	This model type will capture the dataset's underlying patterns (low bias) and isn't very sensitive to the training data changes (low variance). This ML model is ideal as it can produce accurate and consistent predictions. However, it is not possible in practice.

Difference between bias and variance:

Bias	Variance
Bias occurs in a machine learning model when an algorithm is used but does not fit properly.	Variance is the amount of variation the target function estimation will change if different training data is used.
It is the difference between the actual values and the predicted values.	It talks about how much any random variable deviated from the expected value.
The model cannot find patterns in the training dataset, failing for unseen and seen data.	The model can find most patterns from the dataset. It learns from noise or unnecessary data.

Thank You!

